Silent-Failure Study

Status as of 5 September 2026 · **measurement complete**

Where you are: three problem families, three engines, an independent referee that shares no library with any of them.

(A) Engines can report success while returning wrong physics — SUPPORTED. (B) Engines *differ* in whether they do — NOT supported. Every engine behaved identically and correctly on every case measured, in all three families.

The three families

Family	New machinery	Degeneracy	Result
1. Pendulum chain	baseline; 1–30 links	amplitude drives it chaotic ($10^{11}\times$ growth)	all agree
2. Slider–crank	prismatic joint, closed loop	dead centre: $dx/d\theta \to 0$, inertia collapses $10^6\times$	all agree
3. Cart–pole	mixed joints in a tree, coupled $M(q)$	light cart: $\det M$ collapses $10^6\times$ at vertical	all agree

72 engine–case pairs across families 2 and 3: 0 disagreements, 0 refusals, worst residual 1.3×10^{-14}, on derivation *and* integration, including at the degenerate configurations.

Capability: they do *not* differ

The earlier 5 / 12 / 30 ladder measured how each engine was *driven*, not what it can do. Given the best path through each engine's own public interface and an equal wall clock, in one session on one platform:

Equal-budget ladder	SymPy	MechanicsDSL	Drake
Largest answered	**80**	**80**	**80**
Time at $N = 80$	693 s	1101 s	**0.2 s**
Residual at $N = 80$	1.2×10^{-12}	$\mathbf{5.9 \times 10^{-13}}$	9.9×10^{-10}

No engine failed at any rung and none disagreed with the reference. 80 is where the ladder stopped, not where anything broke. **Reach is not a differentiator; speed is.**

The transferable result

! **An error that is exactly 1.00 or exactly 2.00 across every case is a convention mismatch in *your comparison*, not a defect in the engine.** Three independent confirmations: canonical (q, p) state vs. accelerations; Drake's relative joint angles vs. absolute; a mirrored pole axis. Second tell: the error vanishes in a degenerate case (N=1, M=I, θ=0) and grows away from it. Skipping this check would have produced three fabricated findings against widely used software.

Rules still in force

! Engine, scoring path, and the 55-case matrix frozen at `a8dc2b2`. Harness may grow; gate is the frozen matrix still returning 44/0/1/10 bit-identically.

! **Do not claim any reach advantage.** All three engines answer $N = 80$ under an equal budget. On speed Drake beats both symbolic engines by ~5500× at $N = 80$, and SymPy driven well is ~1.6× faster than MechanicsDSL at every rung. The one axis MechanicsDSL leads is accuracy (~2× over SymPy), on one family, and everything is far inside tolerance.

! On the slider–crank Drake runs in *discrete* mode — constrained dynamics are not exposed through the continuous interface — so it cannot share the pinned integrator and is scored on constraint satisfaction. Report the asymmetry; it follows from the interfaces, not from the dynamics.

What is left

✓ **Artefacts deposited** on Zenodo; the concept DOI is in the paper's artefacts line.

✓ **Reach confound found and corrected.** The frozen ladder gave each engine one driving path; equal treatment dissolves the tiers (TR-2026-06).

✓ **Shared failure measured, not asserted.** Divergence from the unstable equilibrium is linear in working precision at $\ln 10/\lambda = 0.735$ s per digit, and survives to 150 digits: structural, not an artefact of binary64.

□ **Write it up.** Null result with strong methodology, a three-tier capability comparison, and the diagnostic above. Short or workshop paper, as `SCOPE.md` always said.

Reports: TR-2026-01 referee · 02 SymPy · 03 Drake · 04 Week 1 findings · 05 three families · 06 driving. **Disclose:** you maintain one of the engines measured; AI assistance was used. Both permitted, concealing either is not.